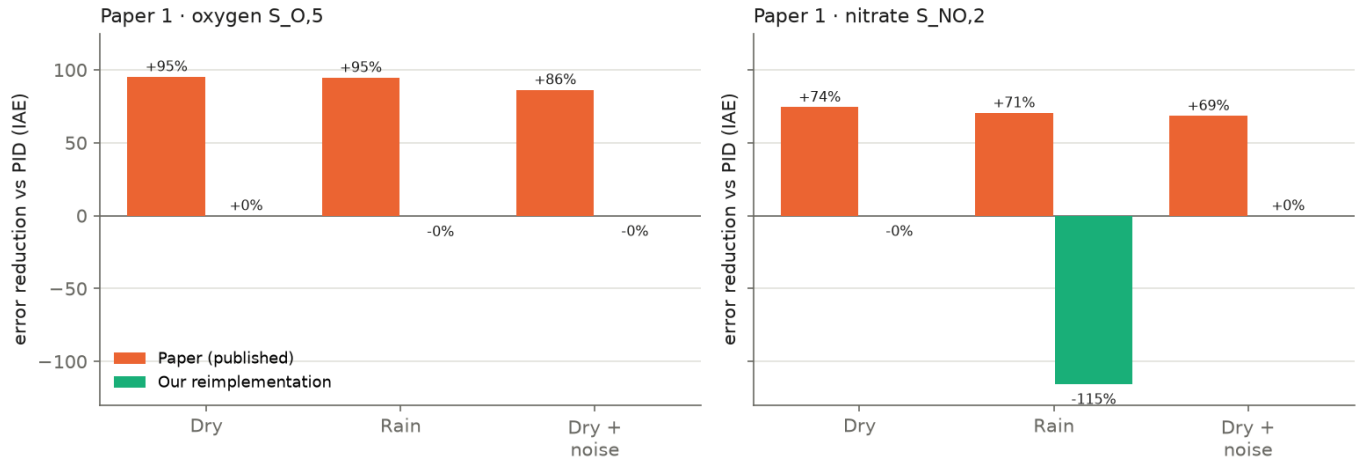
Base paper reimplementation

Two RL control papers, reimplemented on the official BSM1 wastewater benchmark

Paper 1: MAACC — Wang et al., IEEE T-II 2026
Paper 2: floating DO by DDPG — Du et al., Sci China 2023

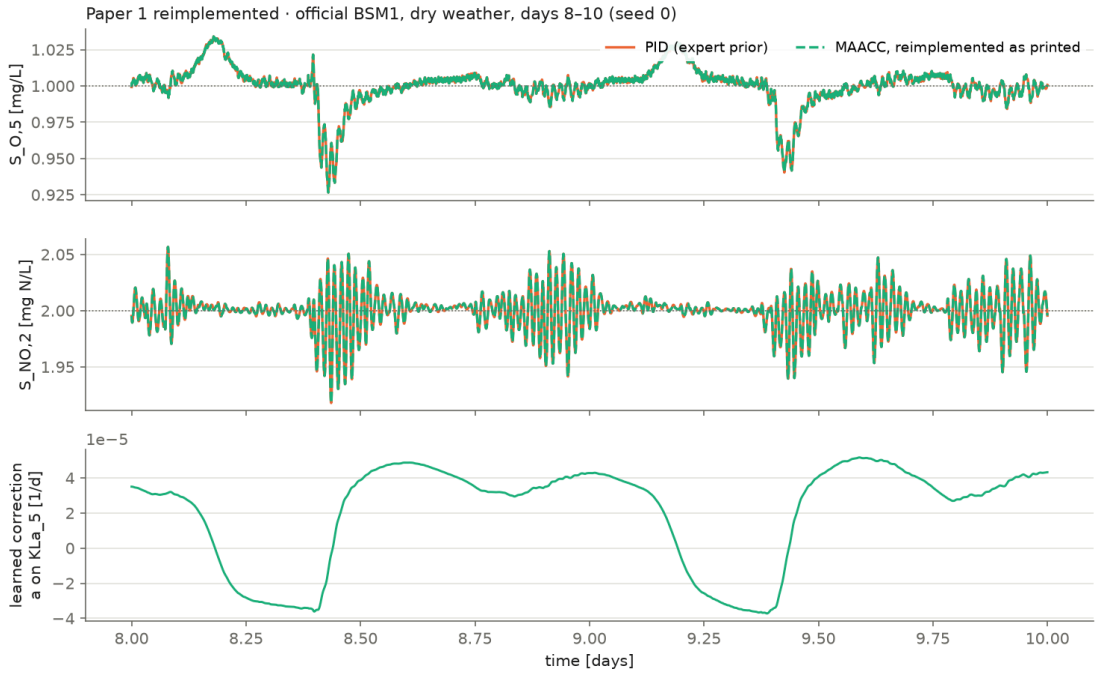
Paper 1 — MAACC

- Paper: oxygen error −95%, nitrate −74% vs PID
- Ours, exactly as printed (Table I, 20 runs): **MAACC = PID** in all 180 runs
- Verdict: **not reproducible from the paper as printed**



Official BSM1 plant and influent, algorithm and Table I values as printed, mean of 20 runs. Up = MAACC better than PID. Reimplemented, MAACC equals PID.

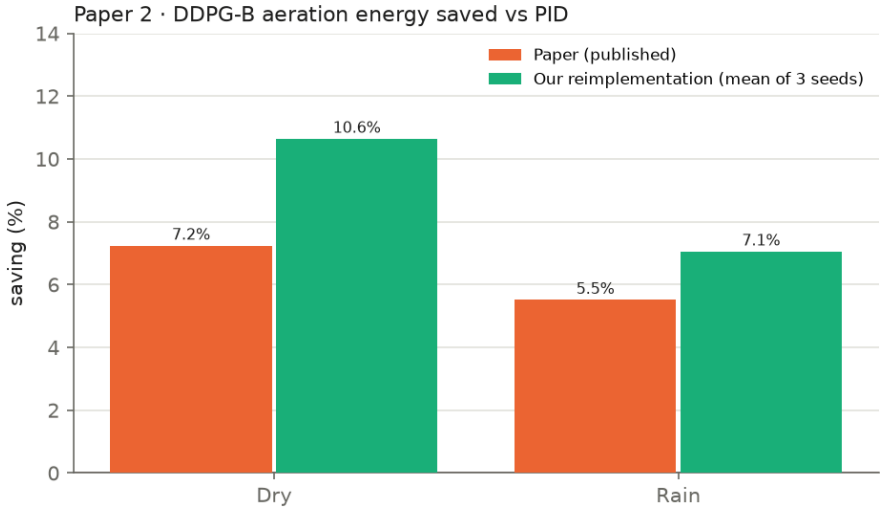
Paper 1 — why



Learned correction stays ~10^{-5}, so MAACC sits on the PID line

Paper 2 — floating DO

- PID & fuzzy match the paper within 1%
- DDPG-B saves 10.6% dry / 7.1% rain (paper 7.2 / 5.5)
- Not in paper: ammonia over limit 56% vs 21% (PID)



Why it could not be reproduced exactly

- **Paper 1 does not learn as printed:** with Table I weights in raw units the cost is $\sim 10^{-4}$, so at $l_c = 0.01$, $l_a = 0.005$ the networks barely move; the correction stays $\sim 10^{-5}$
- **Paper 1's critic update (eq. 15/18) is unstable:** its gain is above 1, so any larger starting weights make it diverge
- **Details missing in paper 1:** starting weights, input scaling, code; 12 starting-weight settings tried, none gives Table II
- **Plant and test setup differ:** no code released; their PID barely changes under the stated noise (ours 3× worse) and does not saturate in rain (ours does); no official storm file
- **Paper 2:** reward energy units, training length and exploration not given; DDPG is seed-sensitive and the paper reports one run, no spread

Conclusion

- Paper 1: not reproducible as printed
- Paper 2: reproduced in direction, size depends on seed
- All results: github.com/Amritha902/water_resource